18.3 - 22 10.12.2026

The Kidneys filiter about 180 liters (~380) of (~380 pints of blood) in an average of an adult a day!

thats about 20% of the total resting volume ⇒ thus serves as ideal sites for receptors that determine O_2 sat = hypoxemia

In response to ~~Hypoexem~~ Hypoexemia, less O_2 will exit the vessels supplying the Kidney, resulting in hypoxia (low O_2) ⇒ tissue fluid of the ~~ff~~ Kidney is where the O_2 ~~is~~ is monitored.

★ Interstitial Fibroblasts within the Kidney secrete ~~EPO~~ EPO ⇒ increasing erythrocyte production & restoring O_2 levels.

In classic Negative - feedback loop, as O_2 saturation rises EPO secretion falls ⇒ vs.

thereby maintaing homeostasis

folks at high altitudes have a higher hematocrit than folks at sea level.

To acclimate / Response to Hypoxemia ⇒ the Kidneys secrete EPO to step up the production of RBCs until homestasis is achieved.

Lifecycle of Erythrocytes

⇒ Production of erythrocytes ⇒

in the marrow occurs @ the staggering rate of more than 2 million cells per second.

raw materials ⇒ ~~glucose~~ glucose, lipids, & amino acids